Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)	$1.00 \sin \theta_{air} = 1.60 \sin 30°$ **or** equivalent **or** by implication (1) $\theta_{air} = 53°$ (1) Beam bends sharply to the left at the surface (1)	1	1 1		3	2	
		(ii)	$\theta = \theta_C$ **or** by implication (1) $\sin \theta = \frac{1}{1.60}$ [=0.625] **or** equivalent (eg θ_C= 39°) (1) x =62.5 / 63 mm (1)	1 1	1		3	2	
	(b)		**Marking points** A1. Light [travelling at small angles to axis] hitting core/cladding boundary is totally internally reflected A2. [So] transmitted along fibre without loss B3. Light paths at different angles [to axis] are of different lengths [for given length of fibre] B4. [So] take [slightly] different times and [so] [each] pulse spread out [over time] on arrival at far end C5. Pulses may overlap (if in rapid sequence) C6. Spreading [accept mm dispersion] increases with length of fibre, [so] overlap more likely if fibre longer **5-6 marks** *Expect at least 4 points made from all of sections A, B and C* *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** *Expect at least 3 points made from at least two sections from A, B or C* *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.*	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			1-2 marks *Any 2 points made* *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						
			Question 7 total	**9**	**3**	**0**	**12**	**4**	**0**